MILITARY SPECIFICATION

WAVEGUIDES, SINGLE RIDGE AND DOUBLE RIDGE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for rigid waveguides with single ridge or double ridge inside configurations (see 6.1).

1.2 Military part number. The military part number shall be as follows:

M	23351/1	- 001	- 096
Military designator	Specification sheet number	Arbitrary dash number (see 3.1)	Length in inches (see 3.3.2)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood Wirebound.
PPP-B-601 - Boxes, Wood, Cleated - Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-66 - Tape: Pressure - Sensitive Adhesive, Vinyl Plastic Film.
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing).
PPP-T-495 - Tube, Mailing, and Filling.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, ATTN: ELEX 5043, Washington D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-131	- Barrier Materials, Watervaporproof, Flexible, Heat-Sealable.
MIL-S-13282	- Silver and Silver Alloys
MIL-F-22191	- Barrier Materials, Transparent, Flexible, Heat-Sealable.
MIL-W-23351/1	- Waveguides, Single Ridge (Bandwidth Ratio 3.6.1).
MIL-W-23351/2	- Waveguides, Double Ridge (Bandwidth Ratio 3.6:1).
MIL-W-23351/3	- Waveguides, Single Ridge (Bandwidth Ratio 2.4:1).
MIL-W-23351/4	- Waveguides, Double Ridge (Bandwidth Ratio 2.4:1).

STANDARDS

FEDERAL

FED-STD-No. 356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with appropriate Test Methods.
MIL-STD-1285	- Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B210	- Aluminum-Alloy Drawn Seamless Tubes.
ASTM B372	- Seamless Copper and Copper-Alloy Rectangular Waveguide Tube.

(Application for copies should be addressed to the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5-1973 - Dimensioning and Tolerancing

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASA B46.1 Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical group and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. The material shall be as specified (see 3.1). Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Aluminum alloy. Aluminum alloys 6061 and 6063 shall conform to the chemical composition limits of ASTM B210. Aluminum alloy 6061 and 6063 waveguides shall be furnished in the F temper.

3.2.2 Copper alloy. Copper alloy shall conform to commercial bronze, 90 percent of ASTM B372.

3 2.3 Copper. Copper shall conform to type OF or DLP of ASTM B372.

3.2.4 Silver alloy. Silver alloy shall conform to the chemical composition requirements for grade C (coin silver) of MIL-S-13282. In addition, silver alloy waveguides shall be furnished in the drawn temper with a Rockwell hardness of 45.0 to 80.0 on the "B" scale.

3.3 Design and construction Waveguides shall be of the design, construction, and physical dimensions specified (see 3.1). Eccentricity (see 6.4) shall not exceed 10 percent. Dimensioning and tolerancing shall be interpreted in accordance with ANSI Y14.5 - 1973.

3.3.1 Finish. The interior and exterior surfaces of the waveguide shall be bright, smooth, dry, and free of scales or oxide.

3.3.2 Length. The finished waveguide shall be furnished in nominal straight lengths with ends, or in specific straight lengths without ends (see 6.2). For waveguides ordered with ends, the nominal length shall be as specified (see 3.1), and the shortest permissible length of the ends shall be 60 percent of the nominal length. The maximum permissible weight of ends shall not exceed 25 percent of the lot weight For waveguides ordered in specific lengths, the length shall be as specified (see 3 1). Waveguides ordered with or without ends shall conform to the tolerances in table I.

TABLE I. Waveguide tolerances.

Length	Tolerance (all plus)
	(inches)
Stock lengths (with ends) - - - -	1
Specific lengths·	
Up to 14 feet, inclusive - - -	1/4
Over 14 feet - - - - - - - - - -	1/2

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3.4 Bow. When measured as specified in 4.6.2, the bow between any two points 2 feet apart on the concave external surface of the waveguide shall not be more than .010 inch on the narrow surface (E-plane) and .020 inch on the wide surface (H-plane), unless otherwise specified (see 3.1). If the waveguide is less than 2 feet long, the allowable bow shall be proportionate to the above requirements.

3.5 Twist. When waveguides are measured as specified in 4.6.3, the maximum twist along the longitudinal axis shall not exceed 1 degree per foot of length on the face of the interior or exterior surfaces.

3.6 Squareness of cut. When measured as specified in 4.6.4, the departure from square of the end of any waveguide shall not exceed 0.010 inch for waveguides up to .625 inch maximum inside waveguide dimension, inclusive, and 0.016 inch-per-inch (one degree) for waveguides over .625 inch maximum inside waveguide dimension.

3.7 Surface roughness. When measured as specified in 4.6.5, the interior surface roughness of the waveguide shall not exceed the values shown in table II.

TABLE II. Surface roughness tolerances of waveguides

Specified major inner dimension (inches)	Allowable surface roughness, maximum arithmetic average (AA)	
	Aluminum and aluminum alloy	Copper, copper alloy, and silver alloy
Up to 4, exclusive - - -	63	32
4 and over - - - - - - -	125	63

3.8 Scratches. When measured as specified in 4.6.6, scratches shall not be more than 0.001 inch deep in the longitudinal direction for waveguides with internal width dimensions of .625 inch or larger. For waveguides with internal width dimensions of less than .625 inch, the depth of scratches shall not exceed .12 percent of the width.

3.9 Marking. Each length of waveguide shall be marked in accordance with method I of MIL-STD-1285, with the part number (see 3.1) and the manufacturer's source code. The manufacturer's name or trademark may also be marked on the waveguide, provided such letters are not expressly forbidden in the contract, and their height does not exceed 50 percent of the outside dimension of the waveguide The markings shall be applied with a permanent ink or paint. Not more than 6-inch intervals shall be permitted from the end of one marking to the beginning of the next marking.

3.10 Workmanship. The waveguide shall be free from defects of a nature that may interfere with normal military applications. The waveguide shall be uniform in composition and wall thickness, straight and smooth from end to end, and shall be free from internal and external mechanical imperfections, and shall have a clean, bright appearance in accordance with good commercial practice. In addition, the interior surface of the waveguide shall be free from burrs, die marks, chatter marks, dirt, grease, scales, and splinters.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.2 Classification of inspections. The inspections specified herein are classified as follows.

a. Materials inspection (see 4.3).
b. Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table III, used in fabricating the waveguides, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE III. Materials inspection.

Material	Requirement paragraph	Applicable specification or publication
Aluminum alloy - - -	3.2.1	ASTM B210
Copper alloy - - - -	3.2.2	ASTM B372
Copper - - - - - - -	3.2.3	ASTM B372
Silver alloy - - - -	3.2.4	MIL-S-13282

4 4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5 1.1 Inspection lot. An inspection lot shall consist of all waveguides of the same part number produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table IV.

TABLE IV. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and Mechanical inspection	3.3 through 3.3.2, 3.9 and 3.10	4.6.1 and 4.6.1.1	1	4

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4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table IV. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table V, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection

TABLE V. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Bow - - - - - - - - -	3.4	4.6.2
Twist - - - - - - - -	3.5	4.6.3
Squareness of cut - -	3.6	4.6.4

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.4), delivery of products which have passed groups A and B shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table VI, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

TABLE VI. Group C inspection.

Inspections	Requirement paragraph	Test method paragraph
Surface roughness - - -	3.7	4.6.5
Scratches - - - - - - -	3.8	4.6.6

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4.5.2.1.1 Sampling plan. The first two lengths of the production lot shall be tested. Thereafter one out of each 25 lengths shall be tested.

4.5.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the lot shall be considered to have failed.

4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity.

4.5.3 Inspection of packaging. Except when commercial packaging is specified (see 6.2), the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.6 Methods of inspection.

4.6.1 Visual and mechanical examination. Waveguides shall be examined to verify that the materials, design, construction, physical dimensions, finish, length, marking, and workmanship are in accordance with the applicable requirements (see 3.2 through 3.3.2, 3.9 and 3.10.

4.6.1.1 Dimensions. Waveguides with 1-inch major outside dimension and over shall be measured at a point at least 1 inch from the end; waveguides with less than 1-inch major outside dimension shall be measured at least 1/2 inch from the end to ascertain that all dimensions are as specified (see 3.1) and that the waveguide is of the specified length (see 3.1). Eccentricity may be measured at any point around the periphery. For the purpose of determining conformance with the dimensional requirements prescribed herein, any measured value outside the specified limiting values for any dimension shall be construed as failing to meet specification requirements.

4.6.2 Bow (see 3.4). Bow is the departure from a straight longitudinal line connecting any two points 2 feet apart on the external surface of the waveguide without the effect of gravity or any other forces. The waveguide shall be so positioned during measurement that gravity will not tend to increase or decrease the amount of bow. Measurements shall be made along any 2-foot portion of the total length.

4.6.3 Twist (see 3.5). The angle of twist of waveguides shall be measured as shown on figure 1 and as follows:

a. The waveguide shall be placed on a flat, horizontal, reference plane. Waveguides with noticeable, flatwise, bow shall be positioned with the waveguide surface on the side of the convex curvature in contact with the reference plane.

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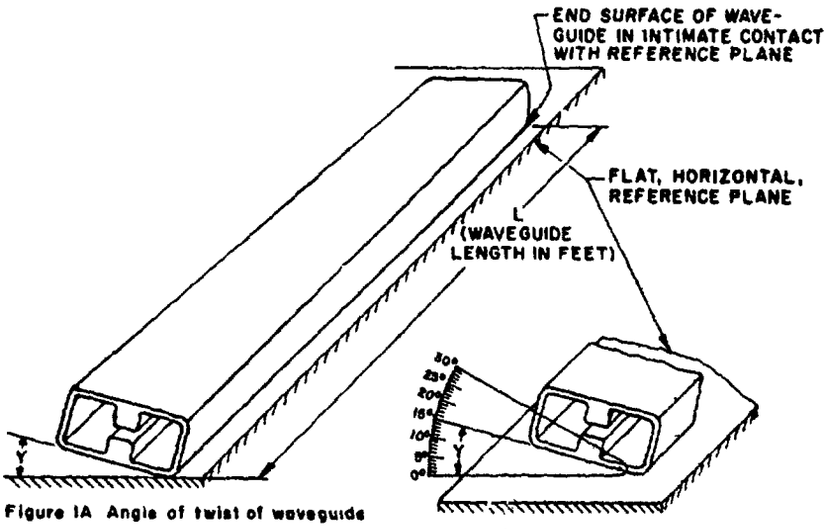


Figure 1A Angle of twist of waveguide

Figure 1B Measurement of total angle of twist

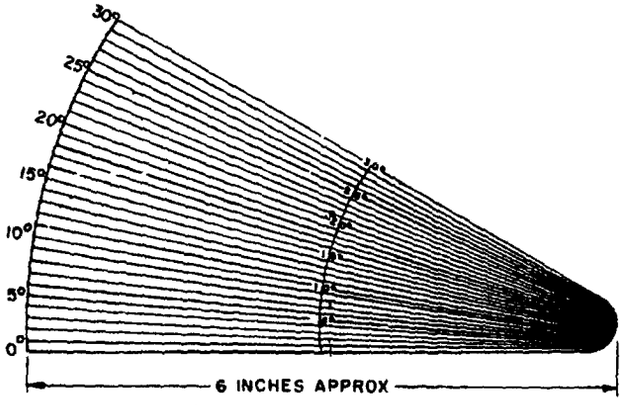


Figure 1C Protractor segment

FIGURE 1. Method of determining twist.

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b. One end of the waveguide surface shall be held in intimate contact with the reference plane by suitable means while the other end of the waveguide remains free.
c. Using an engraved, transparent, protractor segment, determine the angle line on the protractor which coincides to the nearest degree with the surface of the waveguide, while one edge of the protractor segment contacts, or is kept parallel to the reference plane by means of parallel face blocks.

4.6.4 Squareness of cut (see 3.6). The squareness of the cut of the waveguide shall be measured using an adjustable protractor.

4.6.5 Surface roughness (see 3.7). The average interior surface roughness of the waveguide shall be measured in accordance with ASA B46.1, except flaws shall be included in the surface roughness measurement and the corner radius shall not be included in the measurement. Surface roughness shall be measured across the grain over the full width of the waveguide interior surface, at the center of the waveguide length and 6 inches from the end of the length, unless otherwise specified (see 3.1 and 6.2).

4.6.6 Scratches (see 3.8). Inspection for scratches shall be 6 inches from each end and at the center of the waveguide.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Waveguides shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Waveguides shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging.

5.1.1.4.1 Waveguides 36 inches or less in length. Unless otherwise specified (see 6.2), waveguides 36 inches or less in length shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the applicable requirements of that specification. Each unit package shall be placed in a supplementary container conforming to PPP-B-566, PPP-B-676, PPP-T-495, or PPP-B-636 except that waveguides exceeding 12 inches in length shall be restricted to supplementary containers conforming to PPP-B-636.

5.1.1.4.2 Waveguides exceeding 36 inches in length. Unless otherwise specified (see 6.2), waveguides exceeding 36 inches in length shall be individually packaged in accordance with MIL-P-116, method IA as follows: The open ends of each waveguide shall be covered by a water-vaporproof barrier material conforming to MIL-B-131, class 1 or MIL-F-22191, type I. This barrier material shall extend a minimum of 6 inches back from each open end and shall be secured and sealed with tape conforming to PPP-T-66, type I. When the opening exceeds the width of the tape, cross strips shall be used to reinforce the barrier material. Alternately, the openings may be sealed with cover plates or caps and plugs secured with tape conforming to PPP-T-66, type I. All sealing devices shall meet the water-vaporproof requirements of MIL-B-131. The unit container shall conform to the level of packing specified (see 5.2).

5.1.1.5 Intermediate packaging. Not required.

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5.1.2 Level C. Waveguides shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level may conform to the contractor's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, or as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Waveguides 36 inches or less in length. The packaged waveguides shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix of PPP-B-636 using nonmetallic or tape banding only.

5.2.1.2 Waveguides exceeding 36 to 72 inches in length. The packaged waveguides shall be individually packed in a fiberboard container contorming to PPP-B-636, class weather resistant, style optional, special requirements. Blocking and bracing shall be provided as necessary in accordance with MIL-STD-1186 to prevent movement of the waveguide within the container. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix of PPP-B-636 using nonmetallic or tape banding only.

5.2.1.3 Waveguides over 72 inches in length. The packaged waveguides shall be individually packed in a wood box conforming to PPP-B-601, overseas type, grade A or PPP-B-621, class 2, grade A. Blocking and bracing shall be provided as necessary in accordance with MIL-STD-1186 to prevent movement of the waveguide within the container. Closure and strapping requirements shall be as specified in the applicable specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1.1, 5.2.1.2, and 5.2.1.3 except that the container shall conform to the domestic type or class of the respective specification.

5.2.3 Level C. The packaged waveguides shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations and may be the contractor's commercial practice when such meets the requirements of this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Packed waveguides exceeding 52 inches in length shall be unitized by securely banding (strapping) the packs together and when exceeding 200 pounds in gross weight shall have skids applied in a manner that will adequately support the items and facilitate the use of material handling equipment. Waveguides packed in containers not exceeding 52 inches in length shall be unitized in accordance with the applicable procedure specified in 5.2.4.1 through 5.2.4.3."

5.2.4.1 Level A. Waveguides packed as specified in 5.2.1 shall be placed on pallets in conformance with MIL-STD-147, load type 1, with a fiberboard cap (storage aid 4) positioned over the load.

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5.2.4.2 Level B. Waveguides packed as specified in 5.2.2 shall be palletized as specified in 5.2.4.1 except that the fiberboard cap shall be class domestic.

5.2.4.3 Level C. Waveguides packed as specified in 5.2.3 shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carriers rules and regulations.

5.3 Marking. In addition to any special marking required by the contract (see 6.2), each unit package, supplementary and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.3.

5.4.3 Army procurements.

5.4.3.1 Level A (maximum military protection) unit packaging. MIL-P-116 submethod IC-1 and method IC shall be used in lieu of submethod IA-8 and method IA, respectively. All supplementary containers shall either be weather resistant or overwrapped with waterproof barrier materials or shall conform to PPP-B-566 or PPP-B-676, variety 2 or 4 (see 5.1.1.4.1, 5.1.1.4.2, and 6.2).

5.4.3.2 Level A (maximum military protection) and level B (minimum military protection) packing. For level A packing the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner that will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.3.3 Commercial packaging. Commercial packaging (including unit packaging, packing and marking) shall be in accordance with Federal Standard No. 356.

6. NOTES

6.1 Intended use. Waveguides covered by this specification are intended for use as transmission lines in military services radio and electronic equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. The complete part number (see 1.2 and 3.1).
c. Inspection of commercial packaging (see 4.5.3).
d. Levels of preservation-packaging and packing required (see 5.1 and 5.2).
e. Method of preservation, if other than submethod IA-8 or method IA (see 5.1.1.4.1, 5.1.1.4.2, and 5.4.3.1).
f. Special marking, if required (see 5.3).
g. Stock lengths or specific lengths of waveguide required (see 3.3.2).
h. Special methods or increased frequency of checking surface roughness (see 4.6.5).

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6.3 Engineering information. Additional engineering information for waveguides is available in MIL-HDBK-216, "RF Transmission Lines and Fittings," copies of which are available upon request from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Definitions. For the purpose of this specification, the following definition shall apply:

6.4.1 Eccentricity. Eccentricity is the maximum deviation of the wall thickness at any cross section from the mean wall thickness at that cross section.

6.4.2 Specific length. A waveguide furnished to a specified exact dimension, intended for end use in an application which requires a certain length.

6.4.3 Stock length. A waveguide furnished to a nominal length, intended for end use in an application which may require the waveguide to be cut into smaller pieces.

6.4.4 Ends. A section of waveguide resulting from cutting a manufactured piece of waveguide into stock lengths.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, AS
Air Force - 19

Preparing activity:
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